Experiment No: 1 - Creating NumPy Arrays

Aim

To create and manipulate different types of NumPy arrays using various NumPy functions.

Software Required

Python 3.x, NumPy Library, Jupyter Notebook / Google Colab / VS Code

Theory

NumPy (Numerical Python) is a powerful Python library used for numerical computations and multidimensional arrays.

a) Basic ndarray

Program:

```
import numpy as np
arr = np.array([10, 20, 30, 40, 50])
print(arr)
```

Output:

```
[10 20 30 40 50]
```

b) Array of Zeros

Program:

```
import numpy as np
arr = np.zeros((3,4))
print(arr)
```

Output:

```
[[0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]]
```

c) Array of Ones

Program:

```
import numpy as np
arr = np.ones((2,5))
print(arr)
```

Output:

```
[[1. 1. 1. 1. 1.]
 [1. 1. 1. 1. 1.]]
```

d) Random Numbers in ndarray

Program:

```
import numpy as np
arr = np.random.rand(3,3)
print(arr)
```

Output:

Output varies each execution.

e) An Array of Your Choice

Program:

```
import numpy as np
arr = np.array([[5,10,15],[20,25,30]])
print(arr)
```

Output:

```
[[ 5 10 15]
 [20 25 30]]
```

f) Identity Matrix in NumPy

Program:

```
import numpy as np
arr = np.eye(4)
print(arr)
```

Output:

```
[[1. 0. 0. 0.]
 [0. 1. 0. 0.]
 [0. 0. 1. 0.]
 [0. 0. 0. 1.]]
```

g) Evenly Spaced ndarray

Program:

```
import numpy as np
arr = np.linspace(0,100,11)
print(arr)
```

Output:

[0. 10. 20. 30. 40. 50. 60. 70. 80. 90. 100.]

Result

Thus, different types of NumPy arrays were successfully created using NumPy functions.